

CLASSROOM IMPLEMENTATION GUIDE

Scale Quest

From Atoms to Space

尺度大冒險 從原子到太空

PURPOSE A bilingual, game-based sequence for building scale sense before introducing SI-prefix conversion.

Designed for Hong Kong secondary science and physics classrooms

Suggested level: Years 8-10 | Core activity: 15-20 minutes | Individual or pairs

Play the game: <https://scale-quest-hk.ym03chan.chatgpt.site>

Teacher edition | July 2026

1. Quick start

THE BIG IDEA Students first judge and order familiar objects. Only then do they attach powers of ten and prefixes. This protects the activity from becoming symbol matching without scale sense.

Learning outcomes

- Estimate and compare the order of magnitude of familiar, microscopic, geographic and astronomical lengths.
- Use 10^0 m as the central base-unit anchor and distinguish metre (m) from the milli prefix (m in mm).
- Connect the core 10^3 ladder: nano, micro, milli, metre, kilo, mega and giga.
- Convert a length among scientific notation, metres and an appropriate SI-prefix form.

Suggested 18-minute lesson

Time	Student activity	Teacher move
0-2 min	Mission 1: Compare	Ask for a reason before revealing any number.
2-5 min	Mission 2: Size Line-up	Listen for comparison of the named dimension.
5-9 min	Mission 3: Zoom Map	Pause at the classroom door: 2 m belongs near 10^0 m.
9-12 min	Mission 4: Prefix Power-ups	Chant or point through the 10^3 ladder once.
12-17 min	Mission 5: Triple Match	Require both selected cards to be justified.
17-18 min	Exit check / optional Sprint	Use Sprint only after learning, never as the entry task.

PAIRING Recommended roles: Scale Scout estimates the object; Unit Engineer checks the exponent and symbol. Swap roles after Mission 2.

ACCESS STATUS This release is owner-only. Before students or colleagues can open it on their own devices, the site owner must explicitly switch it to public link access.

2. Before class

Device and room setup

- Open the game on student laptops or iPads. One device per pair encourages productive explanation.
- For whole-class demonstration, use Teacher Mode > Projection mode to enlarge classroom text.
- Decide whether to keep the guided locks or enable Open all missions for station work or revision.
- Set the optional Sprint to 45, 60 or 90 seconds. The introductory missions are untimed.
- Progress is stored only on that browser. Reset student progress before a new class if devices are shared.

Read this launch script to students

ENGLISH Work with your partner. First estimate; do not search the answer. Read the named dimension carefully. Explain your reason before either person taps. Your team has three Scanner hints. The timed Sprint is optional.

中文 與同伴合作。先估算 不要搜尋答案。留意題目指定的量度方向 每次點選前先解釋理由。每組有三次「掃描提示」。限時衝刺屬自選活動。

What students should not do

- Do not judge from the on-screen picture size; the illustrations are not drawn to the same scale.
- Do not treat every displayed value as an exact universal measurement. Familiar objects vary; the game uses representative values.
- Do not start with Sprint. Speed before understanding disadvantages thoughtful students and hides misconceptions.

3. Mission-by-mission teaching notes

Mission 1 - Compare

Purpose: activate intuitive scale knowledge without symbolic load.

- Prompt: 'Which measurement is larger, and what real object are you picturing?'
- Watch for students comparing picture size instead of the named dimension: thickness, diameter, height, length or span.
- Do not reveal numbers after a wrong first attempt. Ask students to imagine the objects side by side.

Mission 2 - Size Line-up

Purpose: build a continuous mental line from microscopic to metre scale.

- Ask students to place the two extremes first, then use them as anchors.
- Students may drag, tap two cards to swap, or use arrow buttons; the method should not affect assessment.
- After success, read the revealed values aloud from micrometres to metres.

Mission 3 - Zoom Map

Purpose: connect objects to orders of magnitude before naming prefixes.

- Make 10^0 m physically central in your explanation: it means one metre, with no prefix.
- When an answer is wrong, ask only: 'too large or too small?' The interface gives the same directional feedback.
- The optional Scale Museum extends to atto/femto/pico and tera/peta. Treat it as enrichment, not required content.

CHECKPOINT QUESTION Why is the classroom door near 10^0 m, while a rice grain is near 10^{-3} m? Ask for a factor comparison, not only the prefix names.

3. Mission-by-mission teaching notes (continued)

Mission 4 - Prefix Power-ups

Purpose: attach language and symbols to an already meaningful scale.

- Core ladder: n (10^{-9}), micro symbol μ (10^{-6}), m (10^{-3}), no prefix (10^0), k (10^3), M (10^6), G (10^9).
- Say explicitly: lowercase m may be the milli prefix in mm, but m alone is the metre unit.
- The statement 'each prefix differs by 10^3 ' applies to this selected core ladder, not to every SI prefix. Centi, deci, deca and hecto are outside this ladder.

Mission 5 - Triple Match

Purpose: integrate object scale, scientific notation and prefix form in manageable rounds.

- Students choose only one scientific-notation card and one prefix card for the displayed object.
- If one card is correct, the game says so without identifying it. Ask students to compare the exponent to the prefix.
- After a correct match, require students to read the full equality from left to right.

Optional Scale Sprint

Purpose: fluency check after conceptual work. It is not a ranking tool.

- Use personal bests rather than a class leaderboard.
- Students who need more processing time may skip Sprint without losing access to the core learning.
- A calm, accurate explanation is stronger evidence than a high speed score.

FORMATIVE ASSESSMENT Before students leave, ask them to write one equality such as $24.5 \text{ mm} = 0.0245 \text{ m} = 2.45 \times 10^{-2} \text{ m}$ and name the relevant prefix.

4. Teacher answer key

Compare and line-up

Task	Expected answer	Representative values
Compare 1	HK\$10 coin diameter is larger	24.5 mm vs card thickness 0.76 mm
Compare 2	Human-hair diameter is larger	70 μm vs red blood cell 7.5 μm
Compare 3	Basketball-court length is larger	28 m vs classroom-door height 2 m
Compare 4	Hong Kong Island length is larger	16 km vs Tsing Ma main span 1.38 km
Line-up	hair < card < rice < coin < door	70 μm < 0.76 mm < 7 mm < 24.5 mm < 2 m

Zoom Map

Object	Scale region	Displayed value
DNA helix width	10^{-9} m	2 nm
Human-hair diameter	10^{-6} m	70 μm
Rice-grain length	10^{-3} m	7 mm
Classroom-door height	10^0 m	2 m
Tsing Ma Bridge span	10^3 m	1.38 km
Earth radius	10^6 m	6.37 Mm
Sun diameter	10^9 m	1.39 Gm

PRECISION NOTE These are representative or rounded values chosen to teach scale and conversion. Object dimensions vary, and approximate values should not be interpreted as measurement standards.

5. Anticipated misconceptions

What you may hear	What it reveals	Useful response
'm means milli.'	Unit-prefix confusion	Ask whether m is alone or attached before another m in mm.
' 10^0 is zero.'	Exponent meaning	Return to $10^0 = 1$, so $10^0 \text{ m} = 1 \text{ m}$.
'The bigger picture must be longer.'	Representation bias	Name the dimension and imagine the real objects side by side.
' $70 \mu\text{m}$ is $70 \times 10^{-5} \text{ m}$.'	Prefix-exponent mismatch	Write $\mu = 10^{-6}$ first, then normalize the coefficient.
'Mega and milli both use m.'	Case-insensitive symbol use	Contrast lowercase m with uppercase M; SI symbols are case-sensitive.
'Every SI prefix changes by 1000.'	Overgeneralisation	Clarify that this game uses a selected 10^3 ladder.

Differentiation

- Support: keep students in pairs, allow Scanner hints and ask only for too large/too small before exact conversion.
- Core: require a spoken reason before each choice and one written equality at the end.
- Extension: open the Scale Museum and ask why atto is more familiar in time than in everyday length.
- Language support: students may play in Traditional Chinese while saying English prefix names such as nano, micro and kilo.

Accessibility and fair use

- Do not grade speed. The untimed missions contain the intended learning evidence.
- Keyboard, tap, swap and arrow controls are equivalent ways to complete sorting.
- Avoid colour-only instructions; refer to labels, symbols, positions and feedback text.

6. Assessment and reflection

Three-item exit ticket

1. Put 2 m, 7 mm and 70 μm in ascending order.
2. Complete: 24.5 mm = _____ m = _____ $\times 10^{\text{_____}}$ m.
3. Explain why metre belongs at 10^0 on the scale and why m alone is not a prefix.

Suggested answers: $70\ \mu\text{m} < 7\ \text{mm} < 2\ \text{m}$; $24.5\ \text{mm} = 0.0245\ \text{m} = 2.45 \times 10^{-2}\ \text{m}$; $10^0 = 1$, so one base metre has no prefix.

Pilot feedback to collect

- Which object comparison felt easiest and which felt least believable?
- At which mission did students begin using powers of ten independently?
- Did students confuse metre m, milli m and mega M after the lesson?
- Were any controls difficult on the actual school laptops or iPads?
- Was 15-20 minutes realistic for your class? Record the actual time.

Sharing with colleagues

4. Pilot the private version with one class and collect the feedback above.
5. When ready, change the site from owner-only to public link access. Until then, colleagues cannot open the current private URL.
6. Share two items together: the game link and this Teacher Guide. Tell colleagues that no student login is required and progress stays on the device.
7. Invite colleagues to test one full mission sequence before using Sprint or opening all missions.
8. Use a short version label and date when circulating updates so staff know which release they tested.

RECOMMENDED MESSAGE I am piloting Scale Quest, a bilingual 15-20 minute science game on length scales and SI prefixes. Please try the five guided missions before the optional Sprint, then send me one strength, one confusing point and your device type.

Current game link: <https://scale-quest-hk.ym03chan.chatgpt.site>